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Analysis and Prediction of Emergency Response Times in San Francisco

San Francisco Fire Department

Report 1

Exploratory Analysis and Modelling Proposal

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San Francisco Department of Emergency Management

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02/05/2026

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1. Client requirements

1.1 Summary

The San Francisco Fire Department responds to a high volume of emergency calls across a complex urban environment. Their key performance metric is **response time**, defined as the interval between call receipt and arrival on scene. There is concern that response times may vary across different areas of the city.

The client requested an analysis of response time patterns across multiple dimensions, including time, call type, priority, and geographic areas. The objective is to identify where delays occur, how frequent they are, and whether certain areas systematically underperform. A key focus is distinguishing between operational inefficiencies and differences in call characteristics.

In addition, the client requires a predictive model to estimate response time using only information available at dispatch. This model is intended to support planning and resource allocation.

Finally, the client emphasised the need for a **fairness assessment**, evaluating whether response times and model predictions are equitable across neighbourhoods after accounting for call mix.

1.2 Requirements Clarifications

Response time is defined as the difference between the call receipt timestamp (`received_dttm`) and the arrival timestamp (`on_scene_dttm`). For incidents with multiple units, the first arrival is used.

Records without valid arrival timestamps correspond to incidents where units did not reach the scene and are excluded. Negative or implausible values are treated as data errors and removed.

A delay is defined relative to comparable incidents. A call is considered delayed if its response time exceeds the **80th percentile within its call type group and priority**. This avoids using a single threshold across different incident types.

Only features available at dispatch time are used for modelling, including call characteristics, location, unit information, and time variables. Post-dispatch variables are excluded to prevent data leakage.

The primary modelling task is to predict response time (continuous). A secondary classification task may predict whether a delay occurs.

Fairness is assessed by comparing response times across neighbourhoods after controlling for call type and priority.

External factors such as traffic, weather, and resource availability are not included in the dataset and may limit analysis and model performance.

2. Dataset Analysis

2.1 General Analysis

The dataset contains detailed records of emergency response activity from the San Francisco Fire Department between 2023 and 2025. Each record represents a **unit-level response**, meaning that multiple records may correspond to a single incident when more than one unit is dispatched.

As a result, response times are measured at the unit level rather than the incident level. For incidents involving multiple responding units, the first arriving unit is used to represent the effective response time experienced by the caller. While this approach reflects operational reality, it may underrepresent variability across units within the same incident and should be considered when interpreting results.

The dataset includes over **1 million records**, covering a wide range of incident types, priorities, and geographic areas. This provides a strong basis for analysing response time patterns across operational dimensions.

Response Time Distribution

Response time — defined as the time between call receipt and arrival on scene — is the central metric of interest. After cleaning, approximately **855,000 valid observations** were retained.

The distribution of response times is **highly skewed**, with most calls handled relatively quickly and a small number of extreme values extending the upper range (see Annex 4.1, Figure A1). The median response time is approximately **8 minutes**, while 80% of calls are handled within **15 minutes**.

This distribution highlights the importance of using **percentile-based metrics**, as averages are sensitive to extreme values.

Data Quality Considerations

Several data quality issues were identified. Approximately **22% of records** lack a valid arrival timestamp, corresponding to incidents where the unit did not reach the scene (see Annex 4.3). These records were excluded from response time calculations.

A small number of negative or implausible values were identified and removed. Extreme outliers are present and require careful interpretation.

Operational Drivers of Response Time

Response times vary substantially across operational dimensions.

Call type is one of the strongest drivers of response time. Non Life-Threatening incidents exhibit substantially slower and more variable response profiles than Alarm, Fire, and Potentially Life-Threatening calls (see Annex 4.5, Figure A2). For example, the 80th percentile response time exceeds 23 minutes for Non Life-Threatening incidents, compared with approximately 9 minutes for Alarm calls.

Geographic disparities are also evident across neighbourhoods, battalions, and zip codes. Some battalions consistently exhibit higher delay rates than others, even after restricting the analysis to high-volume operational areas (see Annex 4.7, Figure A4).

Moderate temporal variation exists across both time of day and day of week. While median response times remain relatively stable throughout the day, longer-delay incidents become more frequent during daytime and afternoon operational periods (see Annex 4.8, Figure A5).

These factors are important when interpreting differences in response times across the city.

Defining and Measuring Delays

A global threshold for delays is not appropriate due to variation across incident types. Instead, delays are defined relative to comparable cases. A call is considered delayed if its response time exceeds the 80th percentile within its call type group and priority (see Annex 4.4).

Summary

The dataset provides a comprehensive view of emergency response operations, with sufficient scale and detail to support analysis. While most calls are handled within expected timeframes, response times vary across incident types and locations, motivating further investigation in Section 2.2.

2.2 Business Questions Analysis

The primary objective of this analysis is to understand where delays occur, how frequently they happen, and whether there are systematic disparities in response times across the city.

Where do delays occur?

Using a relative definition of delay, approximately **20% of calls are classified as delayed**, by construction.

Initial analysis shows substantial variation in delay rates across neighbourhoods (see Annex 4.6, Figure A3). Some areas exhibit significantly higher delay rates than others, suggesting potential performance differences across the operational network.

However, these raw comparisons must be interpreted with caution. When restricting the analysis to neighbourhoods with sufficient call volume, a more reliable pattern emerges. Among higher-volume areas, **Lakeshore, Excelsior, Sunset/Parkside, and Bayview Hunters Point** exhibit the highest delay rates (see Annex 4.6, Figure A3).

How do delays vary across operational dimensions?

Response times differ substantially across incident categories. Non Life-Threatening incidents exhibit considerably slower response profiles than emergency-oriented categories such as Alarm and Fire calls (see Annex 4.5, Figure A2). These differences support the use of context-aware delay definitions rather than a single global threshold. They also suggest that predictive modelling performance should be evaluated across operational subgroups rather than solely through aggregate metrics.

Operational disparities are also observed across battalions. After filtering to battalions with sufficient operational volume, delay rates range from approximately 15% to 29% (see Annex 4.7, Figure A4). Battalions B08, B09, and B10 exhibit the highest delay rates, while B02 and B05 perform comparatively better.

Temporal patterns are more moderate. Median response times remain relatively stable across days of the week and throughout the day, although longer delays become more frequent during daytime operational peaks (see Annex 4.8, Figure A5).

Are there systematic disparities across areas?

To assess whether these differences reflect true operational disparities, response times were analysed relative to expected values for comparable incidents.

This adjusted analysis reveals that **some disparities persist even after accounting for call type and priority** (see Annex 4.9, Figure A6). In particular, **Lakeshore, Sunset/Parkside, and Excelsior** remain consistently slower than expected.

Conversely, several high-density neighbourhoods — including **Tenderloin, Mission, and Western Addition** — exhibit response times that are **faster than expected**, indicating that high demand does not necessarily lead to poorer performance.

How frequent and significant are delays?

The magnitude of these disparities is operationally meaningful. In the most affected areas, response times exceed expected values by **over one minute on average** (see Annex 4.9, Figure A6).

At the same time, much of the variation in response times can be explained by differences in incident characteristics, reinforcing the importance of context-aware performance evaluation.

Summary of Findings

- Delays are not uniformly distributed across the city (see Annex 4.6, Figure A3)
- Differences are partially explained by call type and priority, as reflected in the variation of delay thresholds across incident types (see Annex 4.4)
- Some areas remain slower than expected after adjustment (see Annex 4.9, Figure A6)
- High-demand areas can perform efficiently

2.3 Recommendations

Based on the analysis, several recommendations emerge.

Neighbourhoods such as **Lakeshore, Sunset/Parkside, and Excelsior** should be prioritised for further investigation, as they exhibit consistently higher-than-expected response times (see Annex 4.9, Figure A6).

High-performing areas such as **Tenderloin and Mission** should be analysed to identify operational best practices that may be transferable to underperforming regions.

It is recommended that performance monitoring incorporates **context-aware metrics**, such as relative delay definitions, to avoid misleading conclusions from raw comparisons (see Annex 4.4).

Battalions B08, B09, and B10 should be prioritised for further operational review due to their consistently elevated delay rates relative to peer battalions. Additional investigation into staffing, coverage areas, dispatch allocation, and travel constraints may help explain these differences.

Temporal operational peaks observed during daytime periods suggest that resource allocation strategies may benefit from dynamic staffing adjustments during higher-risk hours.

Finally, the development of a predictive model is expected to provide additional value in anticipating delays and supporting planning decisions.

3. Modelling proposal

3.1 Modelling Task

The objective is to develop a model that predicts emergency response time using only information available at dispatch, supporting operational planning and resource allocation.

The target variable is response time in seconds, computed as the difference between call receipt (`received_dttm`) and on-scene arrival (`on_scene_dttm`).

A secondary classification task may predict the probability of delay based on the relative delay definition established in Section 2 (see Annex 4.4).

The model will use dispatch-time features such as call type, priority, location, unit information, and temporal variables.

In addition to overall predictive accuracy, performance should remain reasonably consistent across operational subgroups, including neighbourhoods, battalions, and call type groups, to avoid disproportionately poor predictions for specific areas.

3.2 Modelling Strategy

The task is primarily framed as a regression problem, with an optional classification formulation for delay prediction.

Baseline approaches based on group averages and median response times will be used to establish performance benchmarks. Linear models such as Ridge and Lasso regression will provide interpretable baselines, while tree-based methods including Random Forest and Gradient Boosting are expected to perform better due to their ability to capture non-linear interactions, geographic variability, and high-cardinality categorical effects.

Exploratory analysis suggests that geographic and operational variables are likely to be among the strongest predictors of response time. In particular, substantial variation was observed across battalions, neighbourhoods, zip codes, and call type groups, while temporal features such as hour of day may contribute to predicting longer delays.

Feature engineering will include:

- Extracting temporal variables
- Encoding categorical features
- Grouping sparse categories
- Treating extreme outliers
- Exploring log-transformed targets due to skewness

Only information available at dispatch will be used to avoid data leakage.

Regression models will primarily be evaluated using MAE and RMSE. Because response times are highly skewed, MAE is expected to better reflect typical prediction error, while RMSE will help assess sensitivity to extreme delays.

Classification models will be evaluated using Precision, Recall, F1-score, and AUC.

Performance will also be evaluated across operational subgroups, including neighbourhoods, battalions, and call type groups, to assess whether prediction errors disproportionately affect specific areas.

Model selection will rely on out-of-sample validation, with time-based splitting considered to better reflect real deployment conditions and reduce temporal leakage risk.

Expected Performance

Response time prediction is expected to be challenging due to the high variability and skewness observed in the dataset (see Annex 4.1, Figure A1). While typical incidents exhibit relatively stable response patterns, extreme delays are likely influenced by operational factors not captured in the available data, such as traffic conditions, concurrent incidents, and real-time unit availability.

As a result, predictive performance is expected to be substantially better for common operational scenarios than for rare extreme events. Nevertheless, the proposed modelling approach is expected to provide useful operational estimates and support identification of higher-risk situations.

3.3 Expected Issues and Risks

Key risks identified during exploratory analysis include data quality issues, high response-time variability, sparse feature combinations, and missing external operational variables such as traffic, weather, and real-time unit availability.

Missing or inconsistent timestamps may introduce noise into the target variable and will be addressed through filtering and timestamp validation during preprocessing.

Because response time is highly skewed and contains extreme outliers, robust evaluation metrics such as MAE will complement RMSE, while log-transformed targets may be explored to stabilise variance.

Sparse combinations of geographic and operational features may reduce model stability. Grouping strategies and frequency-based encoding will therefore be considered for high-cardinality variables.

Exploratory analysis also identified meaningful geographic disparities across neighbourhoods, battalions, and zip codes. Model performance will therefore be evaluated separately across operational subgroups to assess whether prediction errors disproportionately affect specific areas.

Finally, using the first-arriving unit as the response metric may not fully capture variability across all responding units within an incident. Results should therefore be interpreted primarily as estimates of effective incident-level response experience.

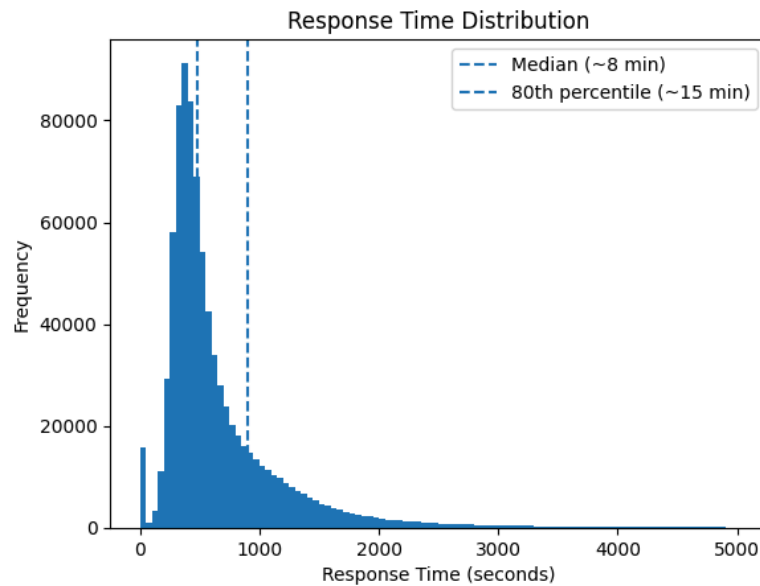
4. Annexes

This section provides supporting analyses and visualisations that underpin the findings presented in Section 2. These materials are included to provide additional detail while maintaining clarity in the main report.

Annex	Content
4.1	Response Time Distribution
4.2	Response Time Summary Statistics
4.3	Data Quality Assessment
4.4	Delay Definition Supporting Analysis
4.5	Response Time by Call Type Group
4.6	Delay Rates by Neighbourhood
4.7	Battalion Delay Rates
4.8	Temporal Response Patterns
4.9	Fairness Analysis (Adjusted Results)

4.1 Response Time Distribution

Figure A1: Distribution of response time (0–5,000 seconds)



The distribution of response times is highly skewed, with the majority of calls occurring within a relatively short time frame and a long tail of larger values. For clarity, the visualisation focuses on response times below 5,000 seconds, which captures the vast majority of observations.

The median response time is approximately **8 minutes**, while the 80th percentile is approximately **15 minutes**, as indicated in the figure. These thresholds form the basis for the delay definition used in the analysis.

Extreme outliers beyond this range exist but are excluded from the visualisation to improve interpretability; they are nevertheless considered in the overall analysis.

4.2 Response Time Summary Statistics

Table A1 summarises key percentiles of the response time distribution.

Table A1: Response Time Percentiles

Percentile	Response Time (seconds)	Response Time (minutes)
50%	485	~8.1
75%	781	~13.0
80%	905	~15.1
90%	1290	~21.5
95%	1700	~28.3
99%	3084	~51.4

These values were used to guide the definition of delays and to contextualise response time performance.

4.3 Data Quality Assessment

Several data quality issues were identified in the dataset.

Table A2: Data Quality Summary

Issue	Count	Approx. Percentage
Missing on-scene timestamps	240,401	~22%
Negative response times	51	<0.01%

A significant proportion of records lack valid on-scene timestamps, indicating that the responding unit did not arrive at the scene. These records were excluded from response time calculations. A small number of records with negative values were identified as data errors and removed.

4.4 Delay Definition Supporting Analysis

Given the variation in response time across incident types and priorities, delays were defined relative to comparable cases.

Table A3 presents example 80th percentile thresholds across combinations of call type group and priority.

Table A3: Example Delay Thresholds (80th Percentile by Call Type Group and Priority)

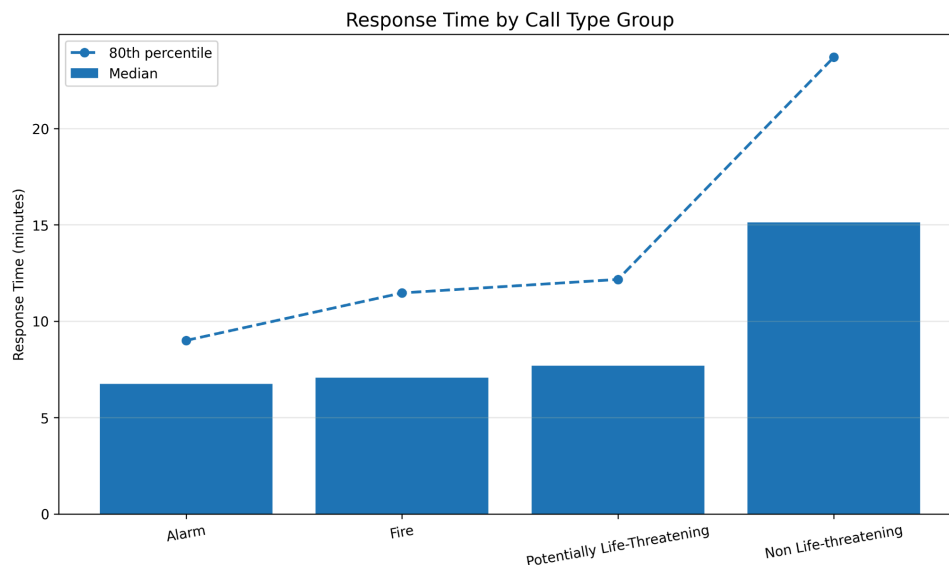
Call Type Group	Priority	Threshold (seconds)
Potentially Life-Threatening	3	~644
Fire	3	~688
Alarm	3	~513
Non Life-Threatening	2	~1383

The wide variation in thresholds across incident types supports the use of a relative delay definition rather than a single global threshold.

Additional operational analysis confirms that response time profiles vary substantially across call type groups. Non Life-Threatening incidents exhibit considerably slower and more variable response patterns than Alarm, Fire, and Potentially Life-Threatening calls. This further supports the use of relative delay thresholds rather than a single global standard.

4.5 Response Time by Call Type Group

Figure A2: Response time by call type group



This figure compares median and 80th percentile response times across major call type groups.

Substantial operational differences are observed across incident categories. Non Life-Threatening incidents exhibit markedly slower and more variable response profiles than Alarm, Fire, and Potentially Life-Threatening incidents.

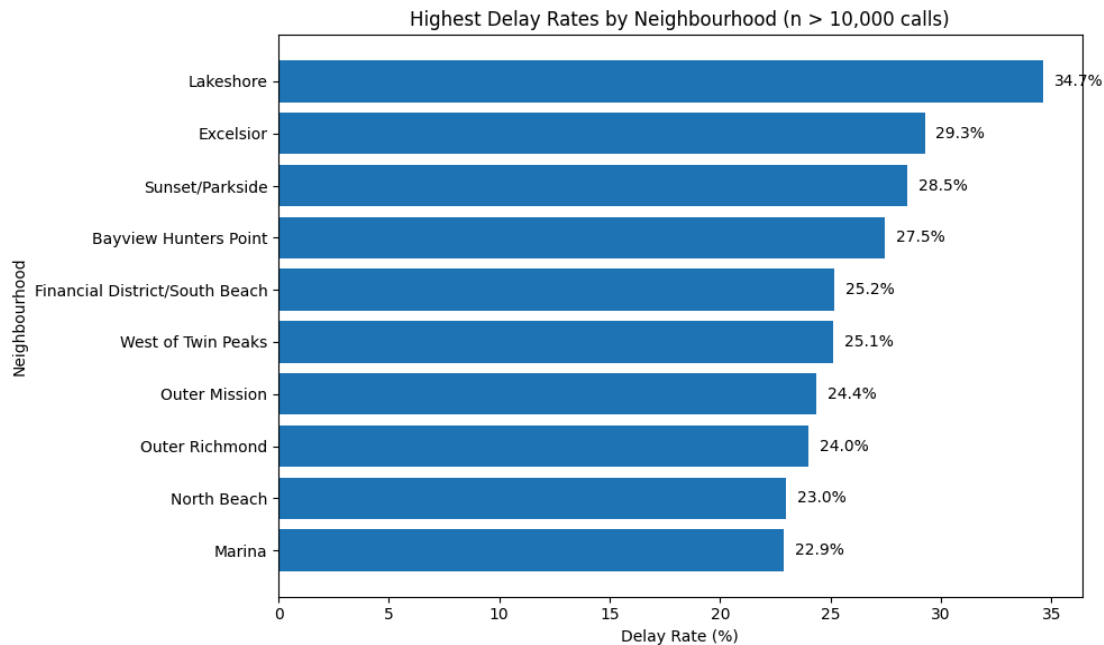
In particular, the 80th percentile response time for Non Life-Threatening calls exceeds 23 minutes, compared with approximately 9 minutes for Alarm calls.

These differences demonstrate that operational expectations vary substantially across incident types and support the use of context-aware delay definitions rather than a single global threshold.

4.6 Delay Rates by Neighbourhood

To ensure reliability, neighbourhoods with fewer than 10,000 observations were excluded.

Figure A3: Highest delay rates by neighbourhood (n > 10,000 calls)



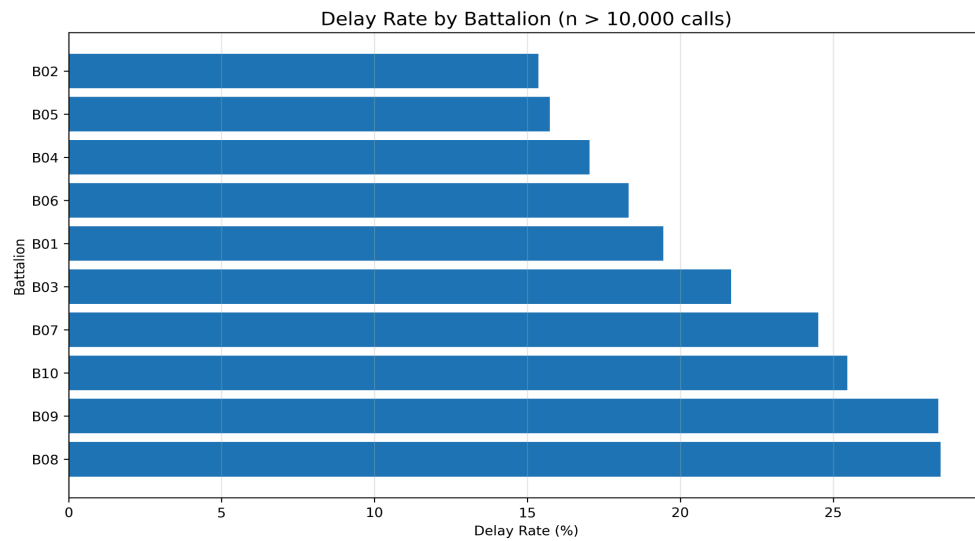
This figure presents neighbourhood-level delay rates after restricting the analysis to areas with more than 10,000 calls. This filtering ensures that the results are statistically reliable and not driven by small sample sizes.

Neighbourhoods such as **Lakeshore, Excelsior, Sunset/Parkside, and Bayview Hunters Point** exhibit the highest delay rates among high-volume areas, confirming that delays are not uniformly distributed across the city.

These findings indicate that geographic disparities are operationally meaningful and warrant further investigation.

4.7 Battalion Delay Rates

Figure A4: Delay rate by battalion ($n > 10,000$ calls)



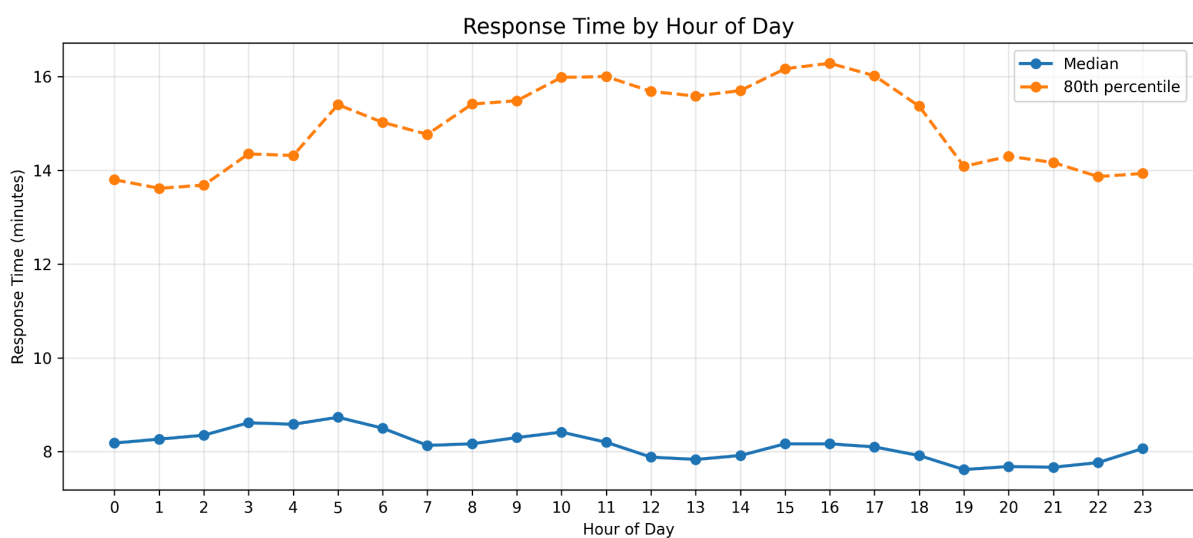
This figure presents delay rates across battalions after filtering to include only battalions with sufficient operational volume.

Substantial variation exists across battalions, with delay rates ranging from approximately 15% to 29%. Battalions B08, B09, and B10 exhibit the highest delay rates, while B02 and B05 perform comparatively better.

These differences suggest that operational disparities are not uniformly distributed across the emergency response network and may reflect differences in geographic coverage, operational constraints, or resource allocation.

4.8 Temporal Response Patterns

Figure A5: Response time by hour of day



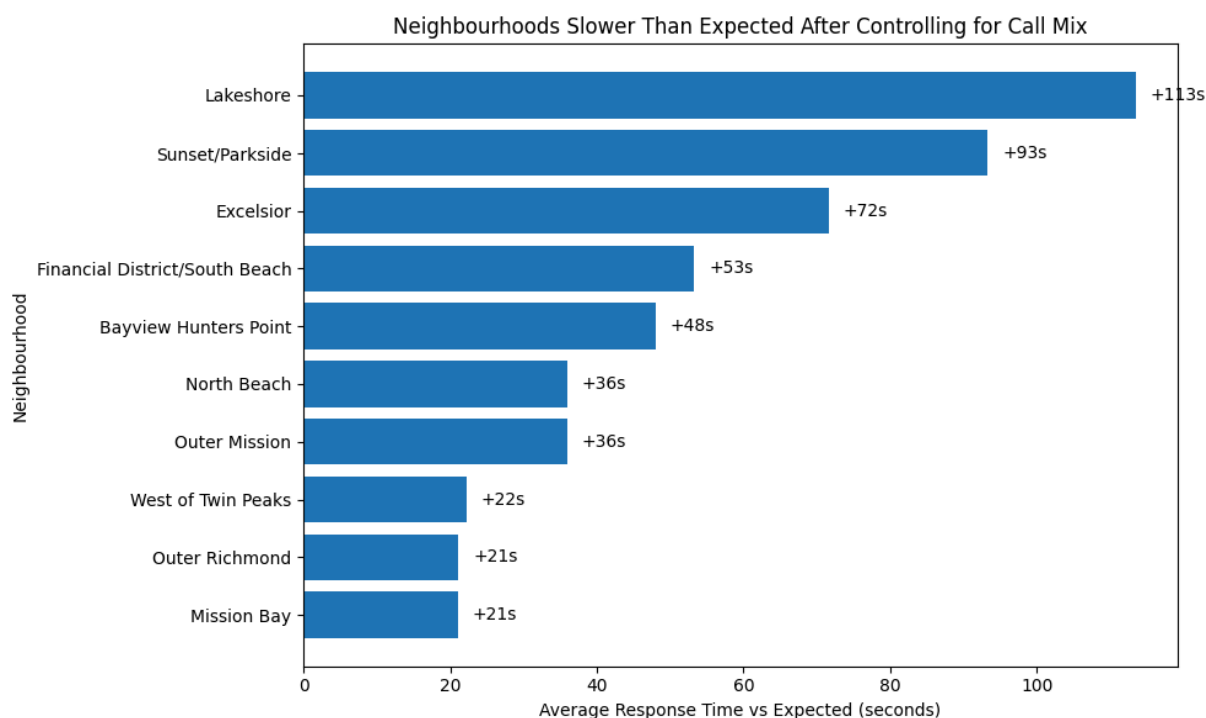
This figure presents median and 80th percentile response times across hours of the day.

Median response times remain relatively stable throughout the day, while the upper tail of the distribution increases during daytime and afternoon operational periods.

The results suggest that operational congestion primarily affects the upper tail of the response time distribution rather than typical incidents.

4.9 Fairness Analysis (Adjusted Results)

Figure A6: Neighbourhoods slower than expected after controlling for call mix



This figure presents the results of the adjusted analysis, where response times are evaluated relative to expected values for comparable incidents (same call type group and priority).

The results show that several neighbourhoods remain slower than expected even after controlling for call mix. In particular, **Lakeshore**, **Sunset/Parkside**, and **Excelsior** exhibit the largest positive deviations.

These findings support the conclusion that some observed disparities are not solely driven by differences in incident characteristics, but may reflect underlying operational or structural factors.

These adjusted results reinforce the importance of evaluating operational performance relative to comparable incident profiles rather than relying solely on raw response time comparisons.